Comment on "Multiobjective model predictive control" [Automatica 45 (2009) 2823–2830]

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Abstract

Lemma 4 of the commented paper states that the value function of the multi-parametric linear program with weight vector μ in the cost and state x in the right-hand side is convex and piecewise affine in μ for fixed x. The function is piecewise affine, but it is *concave* in μ, not convex: for each fixed decision variable the objective is affine in μ and the feasible set does not depend on μ, so the value function is a pointwise minimum of affine functions. We trace the slip to a duality remark in the companion conference paper, and we work out the consequences. The weight-selection linear program of Theorem 6 evaluates the value function as the *maximum* of its affine pieces, so it describes a convex inner approximation of the true admissible weight set, which is in general non-convex: every weight it returns satisfies the contractive constraint, so all closed-loop stability guarantees of the commented paper are preserved, but the claimed equivalence does not hold, and the program can be infeasible while admissible weights exist. In the quadratic case, the joint convexity invoked in Theorem 7 from a 1964 result of Mangasarian and Rosen does not apply, since that result concerns right-hand-side perturbations; the online problem is not a convex program. The practical remedy is to treat the admissible weight set by enumeration or gridding rather than by convex programming.

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1. The claim

The commented paper [1] builds a multiobjective MPC scheme in which, at each sample, a weight vector is selected online subject to the contractive constraint $V^*(\mu, x) \le J_a$, where V^* is the optimal value of a multiparametric program with the weights in the cost. For the piecewise affine case, the machinery rests on [1, Lemma 4], which considers the multiparametric linear problem—their problem (13)—with parameters $\mu \in \mathbb{R}^l$ in the cost function and $x \in \mathbb{R}^n$ in the right-hand side of the constraints, and states that the value function V^* is

> "continuous w.r.t. (μ, x), convex and piecewise affine w.r.t. μ for any given x and w.r.t. x for any given μ."

The statement for x is correct, and so is piecewise affinity in μ. The convexity claim for μ, however, has the wrong sign.

2. The correction

Lemma 1. *Let $V^*(\mu, x) = \min_z\{(c_0 + C'_\mu \mu)'z : \; Gz \le b + Sx\}$ be the value function of problem (13) of [1]. For each fixed x such that the feasible set is nonempty and the minimum is attained, $\mu \mapsto V^*(\mu, x)$ is concave and piecewise affine on its domain.*

Proof. For each fixed feasible z the map $\mu \mapsto (c_0 + C'_\mu \mu)'z$ is affine. The feasible set $\{z : Gz \le b + Sx\}$ does not depend on μ. Hence $V^*(\cdot, x)$ is a pointwise minimum of a family of affine functions of μ, and therefore concave. Since for a linear program the minimum is attained at one of finitely many vertices (or the problem is unbounded), $V^*(\cdot, x)$ is the minimum of finitely many affine functions, hence piecewise affine. $\square$

This is the standard sensitivity dichotomy of parametric optimization: under minimization the optimal value is convex in right-hand-side perturbations and concave in parameters that enter the objective linearly [3, Sec. 5.6.1, Ex. 3.9]. Both signs invert under maximization. The origin of the slip appears to be the companion conference paper [2, p. 2406], which establishes the properties in x and adds that "by duality, the same properties hold with respect to μ." Duality does exchange right-hand-side and cost parameters, but it also turns the primal minimum into a dual maximum, and that is precisely the step that flips the curvature.

We also verified the sign numerically on quadratic instances with the same parametric structure (weights entering the cost linearly, feasible set independent of the weights): over 300 randomly generated chord tests the concavity inequality held in all cases and the convexity inequality failed in all cases, while the same test applied to x confirmed convexity, exactly as in the corrected statement.

3. Consequences within the commented paper

Theorem 6 and the weight-selection LP (17). The proof of Theorem 6 in [1] invokes the result of Schechter [5] for *convex* piecewise affine functions to evaluate $V^*(\mu, x)$, for fixed x, as the *maximum* of the affine pieces $\{(\phi_i x + \gamma_i)'(c_0' + C_\mu' \mu)\}_{i \in I(x)}$, and the linear program (17) accordingly imposes

$$(\phi_i x + \gamma_i)'(c_0' + C_\mu' \mu) \;\leq\; J_a \qquad \text{for all } i \in I(x).$$

By Lemma 1 the value function is the *minimum* of those pieces. Requiring every piece to lie below J_a is therefore sufficient but not necessary for $V^*(\mu, x) \leq J_a$: the constraint set of (17) is a convex *inner approximation* of the true admissible weight set

$$\mathcal{A}(x, J_a) = \{\mu : \; V^*(\mu, x) \leq J_a\},$$

which, being a sublevel set of a concave function, is in general non-convex (for scalar μ it is a union of at most two intervals whenever the maximum of $V^*(\cdot, x)$ is interior and exceeds J_a).

Two things follow, one reassuring and one not. Since every weight accepted by (17) satisfies the contractive constraint, all closed-loop stability guarantees built on that constraint are untouched; the scheme is safe as published. What fails is exactness: (17) is not equivalent to the weight selection problem it is meant to solve, it can be infeasible while $\mathcal{A}(x, J_a)$ is nonempty, and its optimizer need not be the closest admissible weight to the desired one. In our numerical experiments with the corrected sign, the admissible set was non-convex for roughly one fifth of the contraction levels sampled, so the gap is not a measure-zero curiosity.

Theorem 7 and the quadratic case. For the branch with one quadratic objective, the proof of Theorem 7 in [1] asserts joint convexity of $V_\mu^*(\mu, x)$ in (μ, x) "by the results of Mangasarian and Rosen (1964)," and concludes that the online problem (21) is a convex program. In problem (20), however, μ enters the objective linearly through the term $\mu' C_\mu z$ while the constraints $Gz \leq b + Sx$ do not depend on μ; the situation covered by Mangasarian and Rosen [4] is perturbation of the constraint right-hand side, for which the value function of a minimization problem is indeed convex. In the μ direction the same argument as in Lemma 1 applies: $V_\mu^*(\cdot, x)$ is a pointwise minimum of functions affine in μ, hence concave, and joint convexity in (μ, x) cannot hold except in degenerate cases. Consequently the feasible set of (21) is in general non-convex and (21) is not a convex program. Unlike the piecewise affine case, no inner-approximation safety net is created here; any candidate weight returned by a solver should therefore be certified a posteriori by evaluating V_μ^* directly, which is cheap.

Practical remedy. The admissible weight set can be handled by enumeration over the finitely many affine pieces, or by gridding the simplex and testing $V^*(\mu, x) \leq J_a$ pointwise; both preserve exactness at modest cost for the low weight dimensions typical of multiobjective MPC. Projection or gradient steps on the weight, and any construction that presumes convexity of $\mathcal{A}(x, J_a)$, can silently leave the admissible set.

Closing remark

The contribution of the commented paper—a multiobjective MPC formulation with closed-loop guarantees obtained through a contractive constraint on a weighted value function—is not in question, and the published algorithm remains safe precisely because the affected construction errs on the conservative side. The correction concerns the exactness claims attached to the weight-selection subproblems. The authors were contacted about this observation prior to submission.

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